

www.vitrox.com

V810i S2EX Safety



ABI V810i S2EX Safety

Automated Board Inspection



ABI V810i S2EX Certification

Automated Board Inspection

Regulator Marking	Description
	This instrument complies with the WEEE Directive (2002/96/EC) marking requirement. This affixed product label indicates that you must not discard this electrical/electronic product in domestic household waste.
	The CE Mark is a mandatory conformity symbol required for products sold in the European Economic Area. The CE mark indicates conformity to the legal requirements of the European Union (EU) Directive with respect to safety, health, environment, and consumer protection. By affixing the CE marking, the manufacturer, its authorized representatives, or any entity placing the product on the market asserts that the item meets all of the essential requirements of all the applicable EU Directives and that the applicable conformity assessment procedures have been applied correctly.
	TÜV Rheinland is a testing laboratory accredited by the U.S. Occupational Safety and Health Administration (OSHA) and the Standards Council of Canada (SCC). The TÜV Rheinland cTUVus Mark is recognized in jurisdictions across the U.S. and Canada as a reliable means of demonstrating compliance with all applicable codes, standards and regulations. Products bearing a TÜV Listed Mark have been independently confirmed to meet the minimum requirements of U.S. and Canadian electrical safety standards.



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Radiation Containment Design

- Classified as Cabinet X-ray System
- U.S. Federal Regulations 21 CFR 1010.2, 1010.3 & 1020.40
- Radiation emission less than 0.05mR/hr outside the cabinet X-ray system when X-rays are on
- Customers need to comply with local / state / country regulations

❖ Motion Mechanics

- The motion mechanics of the V810i S2EX system involve high speeds and energies.
- Motion mechanics include:
XY Stage assembly – if any access panel is open, the motion components are disabled, preventing any movement of the stage.

ABI V810i Safety Features

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❖ Shielding

- X-ray cabinet using Lead material (3.175mm and 4.76 mm thickness) + Electro Galvanized (EG) to conceal the lead material.
- The inner barrier shutter plate using TUNGSTEN ALLOY.
- The main cabinet, inner barrier and all Outer barrier designed with overlapping layer of lead to ensure No leakage of X-Ray sources.

ABI V810i Safety Features

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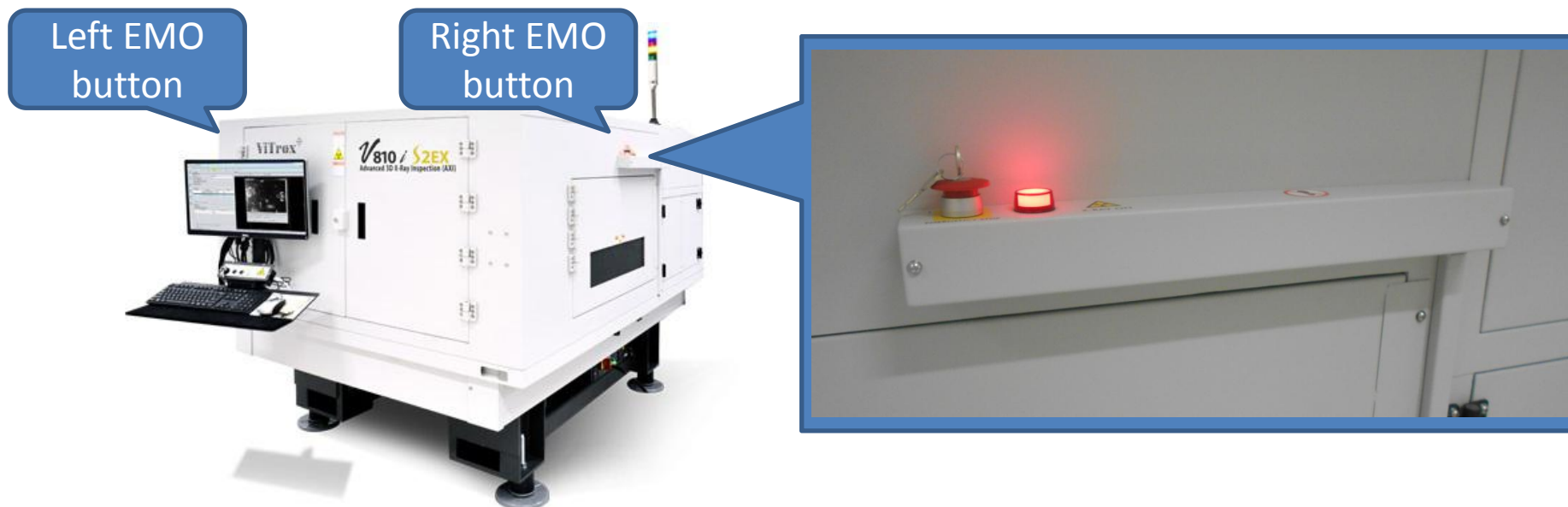
❖ **Safety Interlock Switches**

- To shutdown X-ray sources when any of interlock switches in the V810 system are open.
- Monitoring the status of interlocks and providing enable signal to system (X-ray on and motion enable signal).
- Double redundancy (Chain 1 and Chain 2) is embedded into V810 system as safety measure.



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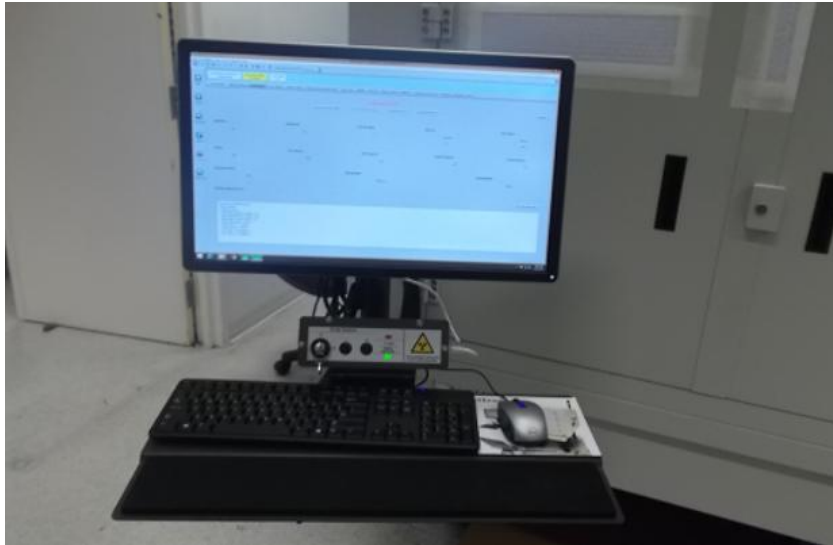
V810i Emergency Stop Switches



- Located on the V810i Main Cabinet
- Emergency power off for switched circuit
- When press, does not power off the (Super Micro) Master Controller, Atom PC, Gigabit Switch, Server Cooling Fans due to PDU configuration
- A key is required to reset the switches if wish to release lock.

ABI V810i Operator Console

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- Located on the monitor workstation
- Turn on X-ray tube ready status

- DO NOT drill holes in the Main Cabinet
- DO NOT place large magnets on the Main Cabinet.
 - Any magnets stronger placed immediately at the position of the deflection coils could cause the electron beam to hit the side of the X-ray Tube.
- DO NOT alter safety interlock switches.
- DO NOT attempt to bypass the safety interlocks
- DO NOT substitute shielding

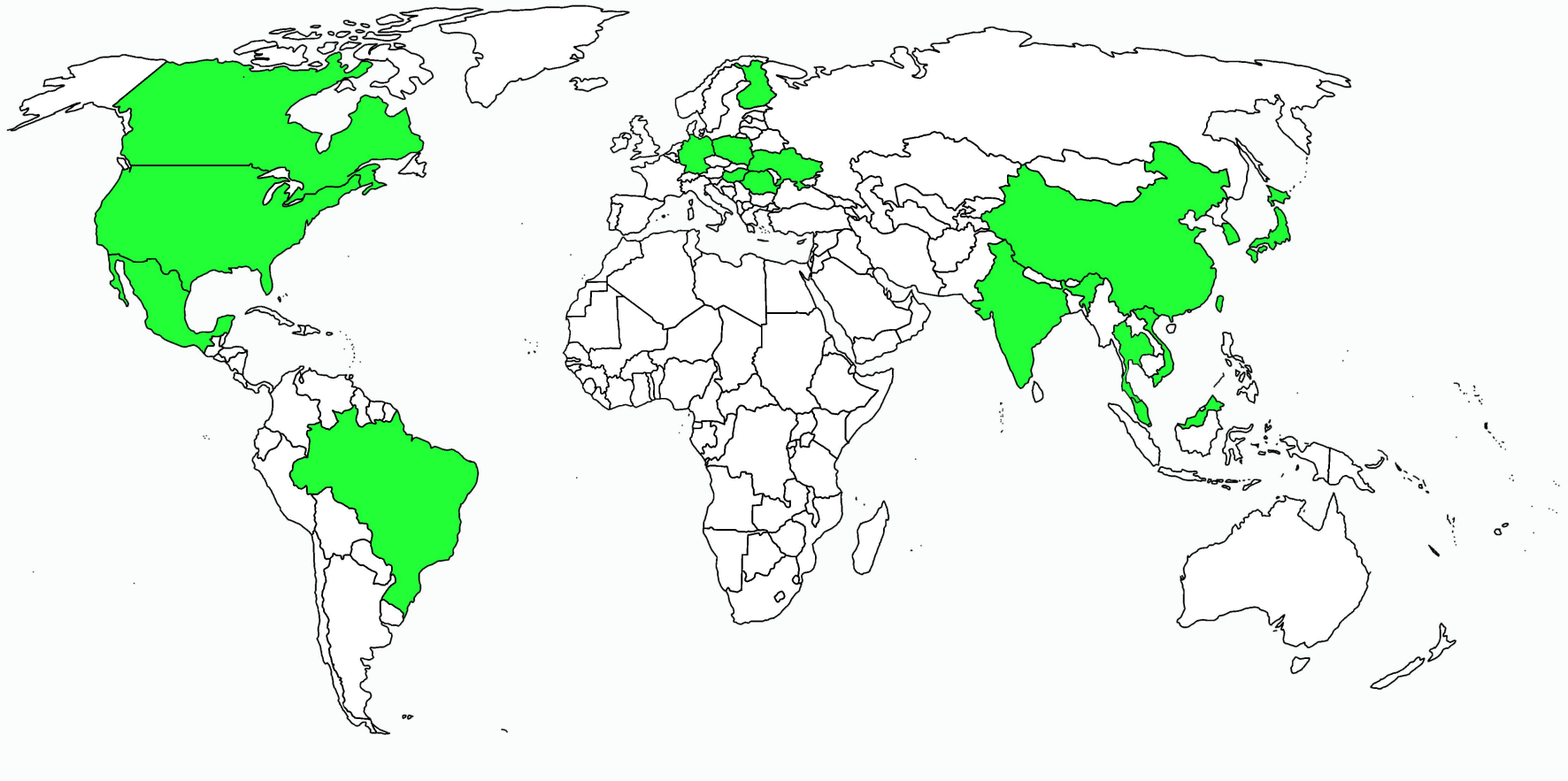
- Background radiation
 - Occurs naturally from the sun, from outer space and natural materials
- Man-made radiation
 - Medical X-rays
 - Nuclear power plants
- Radiation exposure
 - Part of everyone's daily life



**RADIATION
SAFETY**

ABI V810i X- Ray Radiation Specification

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V810i S2EX Spec = 0.05 mR/Hr or 0.5 μ Sv/Hr or 50 μ R/hr

ABI Dose Limit

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- Maximum limit is 5.0 REM per year (CDRH)
- REM is the equivalent measurement of dose exposure and absorption
- V810i S2EX maximum allowable emission is 0.05mR/hr

It caused cells damage, mutate, death

Deterministic Effects

- occur once a threshold of exposure has been exceeded
 - the severity increases as the dose of exposure increases
- (Example: cataract, skin erythema, epilation, sterility, radiation sickness, fetal death and etc)

Stochastic Effects

- have no threshold dose
 - the severity is not dose related
- (Example: cancer, hereditary defects)

ABI Radiation Protection

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Objective

Protection of man and his environment from unnecessary radiation exposure without eliminating the beneficial application of radiation and radioactive materials.

Aim

Prevent deterministic effects and limit the probability of stochastic effect to acceptable level

Vitrox require to perform X-Ray survey during

- Monthly PM
- Semi annual PM
- Change part of inner barrier or outer barrier
- Re-position of inner barrier or outer barrier sensor



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Sample of indicator to measure X-ray Radiation



Optically stimulated luminescence (OSL) Badge



G-M Meter



Survey Meter



Digital Dosimeter

Preparation before perform X-Ray survey

Preparation before perform X-Ray survey :

Step Action

1 Visually inspect the interior and exterior of the V810i system cabinet for any damaged or missing shielding, seals, safety wires or safety screws.

If any are found:

- Inform your manager/supervisor and AXI System Hardware Support.
- Perform a spot test of the affected area using Ion Chamber. Stop system operation if the net emission > 0.05mR/hr (0.5 μ Sv/hr)
- Repair or replace any parts that are damaged or missing.

2 Verify that all X-ray related service notes have been performed on this V810i system:

Select the following X-ray Safety Test:	When doing the following:	Which will take you through the following sections of this procedure:
Comprehensive Test	◆ System installation, ◆ System move ◆ System recertification, or ◆ "First Time" power up in V810i Production	◆ Preparation ◆ Low Power Test ◆ High Power Test
Semi-annual Test	◆ Semi-annual X-ray Safety Test	◆ Preparation ◆ High Power Test
Replaced X-ray tube	◆ X-ray tube replacement	◆ Preparation ◆ High Power
Replaced Lead Shield Test	◆ Repair, replacement or adjustment of Lead Shielding	◆ Preparation ◆ Low Power Test of repaired Lead shielding ◆ High Power Test of repaired Lead shielding Note: Perform the test on the affected area or assembly and the area 12 inches (30.5 cm) out from edge of the affected assembly or area.
Replaced Inner Barrier Test	◆ Repair, replacement or adjustment of inner barrier	◆ Preparation ◆ Low Power Test of opened outer barriers ◆ High Power Test of opened outer barriers
Low Power Test	◆ 130KV and 150uAmps	◆ Low Power Test
High Power Test	◆ 130KV and 300uAmps	◆ High Power Test



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Definition Testing and requirements

Low Power Test 1 : The inner barrier opened and the outer barriers closed.(The Low Power Test is done at 130 kV and 150 μ A.)

Low Power Test 2 : The inner barrier closed and the outer barriers opened. (The Low Power Test is done at 130 kV and 150 μ A.)

High Power Test 1 : The inner barrier opened and the outer barriers closed(The High Power Test is done at 130 kV and 300 μ A.)

High Power Test 2 : The inner barrier closed and the outer barriers opened.(The High Power Test is done at 130 kV and 300 μ A.)

Technique	P-8 Neon	Ion Chamber
Parallel Scanning	Make one pass of the V810i side with the meter. Then move the meter 1.5 in (3.8 cm) and make another pass of the V810i side.	N/A
Finding the Angle of Emission	N/A	The meter must be turned to various angles to the system to find the point and angle of X-ray emission. This angle may be the straight-line path from the X-ray tube to the point of emission.
Scan Rate	2.0 in/sec (5cm/sec)	5 second dwell time (Hold the meter still for 5 sec.)
Meter Clearance	Maintain a gap of between 0.0 in (0 mm) and 0.4 in (10mm) between the meter and the surface under test.	

ABI X-Ray Survey Form

Automated Board Inspection



Model Name : V810i

Model Series: S2EX

Machine SN: _____

Test Date: _____

Customer: _____

Survey Report Type: *Place a check next to the purpose for this survey below.*

- ☐ System Installation ☐ X-ray Tube Replacement ☐ System Re-certification
- ☐ System Move ☐ Repair, replacement or adjustment of Lead Shielding or Inner Barrier (Spot Test) ☐ "First Time" power up in Production
- ☐ Semi-annual Safety Test ☐ Service Note: _____

Ion Chamber *Enter the serial number of the ION Chamber that will be used:*

☐ Victoreen 451P ☐ Others: _____

Meter Serial Number: _____

Probe Serial Number: _____

Calibration Due Date: _____

X-ray Emission

Enter the highest ion chamber reading found during this X-ray Safety Test and compute the Net X-ray emission.

1	Highest X-ray emission reading	Reading	Unit (select one)
2	Background radiation level	Reading	☐ mR/hr or ☐ μ Sv/hr
	Net X-ray emission (1-2)	Reading	

****Note spec < 0.05 mR/hr or 0.5 μ Sv/hr**

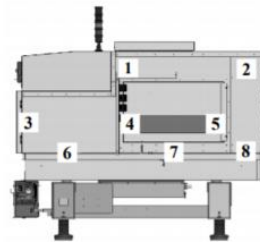
Shut down system and report to ViTrox if out of spec.

Please send completed reports to:

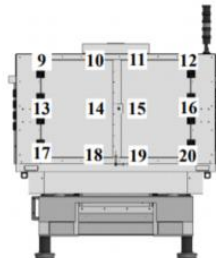
Radiation Safety & Regulatory Engineer
ViTrox Technologies Sdn Bhd
No 85A, Lintang Bayan Lepas 11
Bayan Lepas Industrial Park, Phase 4
11900 Penang
Malaysia

FR-QLT-020 (Rev 01)
Effective Date : 12 October 2016

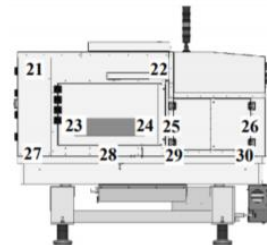
Page 1 of 3



Location	High Power Test 1	High Power Test 2
1		
2		
3		
4		
5		
6		
7		
8		
Max Value		



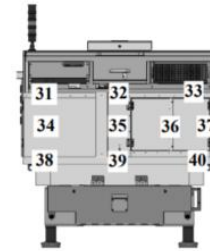
Location	High Power Test 1	High Power Test 2
9		
10		
11		
12		
13		
14		
15		
16		
17		
18		
19		
20		
Max Value		



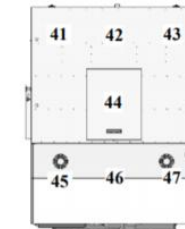
Location	High Power Test 1	High Power Test 2
21		
22		
23		
24		
25		
26		
27		
28		
29		
30		
Max Value		

FR-QLT-020 (Rev 01)
Effective Date : 12 October 2016

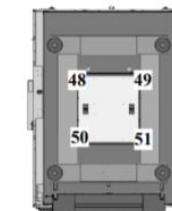
Page 2 of 3



Location	High Power Test 1	High Power Test 2
31		
32		
33		
34		
35		
36		
37		
38		
39		
40		
Max Value		



Location	High Power Test 1	High Power Test 2
41		
42		
43		
44		
45		
46		
47		
Max Value		



Location	High Power Test 1	High Power Test 2
48		
49		
50		
51		
Max Value		

Measured by: _____

Reviewed/Approved by: _____

Name: _____
Date: _____

Name: _____
Date: _____

FR-QLT-020 (Rev 01)
Effective Date : 12 October 2016

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ABI X-Ray Survey - Start

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v810 Service - X-ray Safety Test - 5.11.1 (10706) (Built 07/26/17 at 11:18:05)

File Edit Service Tools Actions Window Help

(kV: 0.0 uA: 0.0)

Confirmation, Diagnostics, & Adjustments Subsystem Status & Control **X-ray Safety Test**

1 **Services**

2

X-ray Source	Panel Positioner	Rail Width	Magnification	Inner Barrier	Outer Barriers
Voltage (kV) 0.0 Current (uA) 0.0	X-Axis (mils) 9000.0 Y-Axis (mils) 11000.0	Width (mils) 19038.8	Low Mag	Closed	Left Closed Right Closed

ViTroX v810 X-ray Safety Test Procedure

Please refer to the ViTroX v810 X-ray Safety Test Procedure document for detailed safety, regulatory and operation instructions.

Purpose
This procedure provides instructions to check the operation of the safety interlocks and to measure the X-ray emission at all points outside the external surfaces of the v810 AXI system cabinet.

Training Requirements
Attending the ViTroX v810 CE Training Class is required before performing the v810 X-ray Safety Test.

Procedure and Report Usage
According to the X-ray Safety Test Procedure, an X-ray Safety Test Report must always be completed. Be sure you have the most recent copy of the ViTroX v810 X-ray Safety Test Procedure and X-ray Safety Test Report.

Pass Specification
Readings must not exceed 0.05 mR/hr (0.5 uSv/hr) when measured with the ion chamber. Any confirmed reading above 0.05 mR/hr (0.5 uSv/hr) shall be considered a failure of the cabinet X-ray system.

Required Tools
Note: Review the ViTroX v810 X-ray Safety Test Procedure document for the most up to date tool requirements.
ViTroX v810 X-ray Safety Test Report
A Geiger-Mueller (GM) meter
An ion chamber

Click Next to select a safety test.

3 **Next**

ABI X-Ray Survey - Select Test Method

Automated Board Inspection

v810 Service - X-ray Safety Test - 5.11.1 (10706) (Built 07/26/17 at 11:18:05)

File Edit Service Tools Actions Window Help

Confirmation, Diagnostics, & Adjustments Subsystem Status & Control X-ray Safety Test

Main

Recipes

Production

Options

Services

Virtual Live

X-ray Source

Voltage (kV) 0.0

Current (µA) 0.0

Panel Positioner

X-Axis (mils) 9000.0

Y-Axis (mils) 11000.0

Rail Width

Width (mils) 19038.8

Magnification

Low Mag

Inner Barrier

Closed

Outer Barriers

Left Closed

Right Closed

Select the type of X-ray Safety Test you would like to run:

- Comprehensive Test
- Semi-annual Test
- Replaced X-ray Tube or HVPS Test
- Replaced Lead Shield Test
- Replaced Inner Barrier Test
- Low Power Test
- High Power Test**
- High Mag High Power Test

1

Test Description

Perform this test if you need to run an X-ray safety test at high power (130 kilovolts, 300 microamps).

Click Next to begin the selected test.

Cancel Back Next Finish

2

ABI X-Ray Survey - Inner Barrier Open

Automated Board Inspection

v810 Service - X-ray Safety Test - 5.11.1 (10706) (Built 07/26/17 at 11:18:05) - Step 1 of 3

File Edit Service Tools Actions Window Help

Confirmation, Diagnostics, & Adjustments Subsystem Status & Control **X-ray Safety Test**

Main

Recipes

Production

Options

Services

Virtual Live

X-ray Source

Voltage (kV) 130.0

Current (µA) 300.0

Panel Positioner

X-Axis (mils) 9000.0

Y-Axis (mils) 11000.0

Rail Width

Width (mils) 19038.8

Magnification

Low Mag

Inner Barrier

Open

Outer Barriers

Left Closed

Right Closed

High Power Test 1

Use these steps to test the system at high power with the inner barrier opened and the outer barriers closed.
A high power test is done at 130 kV and 300 µA.

WARNING If any reading exceeds 0.05 mR/hr (0.5 µSv/hr) when measured with the ion chamber, click **Shut Down** or **Cancel** to bring down the X-ray Current to 0 µA. Follow the process in the X-ray Safety Test Report titled **What if the Reading Exceeds ViTrox's Specification**.

Shut Down

1. To check for major X-ray emissions, scan all cabinet surfaces using an approved Geiger Mueller (GM) meter as stated in the **X-ray Safety Test Report**.
 - If the meter reacts to an area, slow down the scan rate and rescan the area to isolate the location.
2. If the GM Meter reading exceeds 0.1 mR/hr or 1.0 µSv/hr
 - Place a colored paper label on the location of the reading on the system. See the **ViTrox v810 X-ray Safety Test Procedure** document for examples.
 - Set the Geiger Mueller (GM) Meter to the next higher scale.
 - Recheck the location.
 - If the GM Meter exceeds the higher setting, immediately verify the reading with the ion chamber.
 - If not, turn the Geiger Mueller (GM) Meter sensitivity switch back to the most sensitive scale and continue scanning.
3. When the entire system has been scanned with the GM meter, re-check all marked locations with an approved ion chamber. As stated in the **X-ray Safety Test Report**.
4. Write the value of the ion chamber readings (including the units) on the **X-ray Safety Test Report**.
5. Measure six locations on the v810 using the ion chamber. Record the highest reading in that area. There is one location per side shown on the **X-ray Safety Test Report**. Write the highest reading from each of these locations in the **X-ray Safety Test Report**.
6. After the entire exterior of the cabinet has been scanned, click **Next** to go to the next step.

Click Next to continue or Cancel to shut down and exit.

Cancel Back **Next** Finish

ABI X-Ray Survey - Outer Barrier Open

Automated Board Inspection

v810 Service - X-ray Safety Test - 5.11.1 (10706) (Built 07/26/17 at 11:18:05) - Step 2 of 3

File Edit Service Tools Actions Window Help

Confirmation, Diagnostics, & Adjustments Subsystem Status & Control **X-ray Safety Test**

X-ray Source	Panel Positioner	Rail Width	Magnification	Inner Barrier	Outer Barriers
Voltage (kV) 130.0 Current (μA) 300.0	X-Axis (mils) 8999.8 Y-Axis (mils) 11000.0	Width (mils) 19038.8	Low Mag	Closed	Left Open Right Open

High Power Test 2
Use this step to test the system at high power with the inner barrier closed and the outer barriers opened. A high power test is done at 130 kV and 300 μA.

⚠ If any reading exceeds 0.05 mR/hr (0.5 μSv/hr) when measured with the ion chamber, click **Shut Down** or **Cancel** to bring down the X-ray Current to 0 micro amps. Follow the process in the X-ray Safety Test Report titled What if the Reading Exceeds ViTrox's Specification.

Shut Down

1. Rescan the outer barriers using an approved Geiger Mueller (GM) meter as stated in the **X-ray Safety Test Report**.
2. When the outer barriers have been scanned with the GM meter, re-check all marked locations with an approved ion chamber as stated in the **X-ray Safety Test Report**.
3. Write the value of the ion chamber readings (including the units) on the **X-ray Safety Test Report**.
 - After the outer barriers have been scanned, click **Next** to go to the next step.

Click Next to continue or Cancel to shut down and exit.

Cancel Back **Next** Finish



X-Ray Survey - Complete

Automated Board Inspection

v810 Service - X-ray Safety Test - 5.11.1 (10706) (Built 07/26/17 at 11:18:05) - Step 3 of 3

File Edit Service Tools Actions Window Help

Confirmation, Diagnostics, & Adjustments Subsystem Status & Control **X-ray Safety Test**

X-ray Source	Panel Positioner	Rail Width	Magnification	Inner Barrier	Outer Barriers
Voltage (kV) 130.0 Current (µA) 300.0	X-Axis (mils) 8999.8 Y-Axis (mils) 11000.0	Width (mils) 19038.8	Low Mag	Closed	Left Closed Right Closed

The selected X-ray Safety Test has been completed

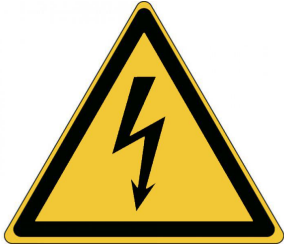
1. Complete the **X-ray Safety Test Report**.
2. Send the completed **X-ray Safety Test Report** to the AXI Hardware Support Manager. The address for the AXI Hardware Support Manager is located in the **X-ray Safety Test Report**.
Note: If this safety test was performed as part of the production process at the ViTrox v810 production facility, store the **X-ray Safety Test Report** in the System Log Book.

Click Finish to return to the initial page.

Cancel Back Next **Finish**

ABI System Warning / Caution Labels

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- ❖ **Electrical Hazard Signs**, indicate High Voltage within.
Allocated: PDU, SSCA/MCA Box, Server Controller (Computers)



- ❖ **General Warning Signs**
Allocated: PDU, Server Controller (Computers)



- ❖ **Do Not Step Signs**
Allocated: Server Rack Top Cover, Top Access Service Door, SSCA/MCA Box, EMO Assembly Mounting Kit (Left and Right), Filter Height Assembly Kit

ABI System Warning / Caution Labels

Automated Board Inspection



❖ X-ray On/Radiation Signs

Allocated: EMO Assembly Mounting Kit (Left and Right)



❖ Mechanical Rail Clamp Movement Signs

Allocated: Fix and Adjustable AWA Conveyor Rail

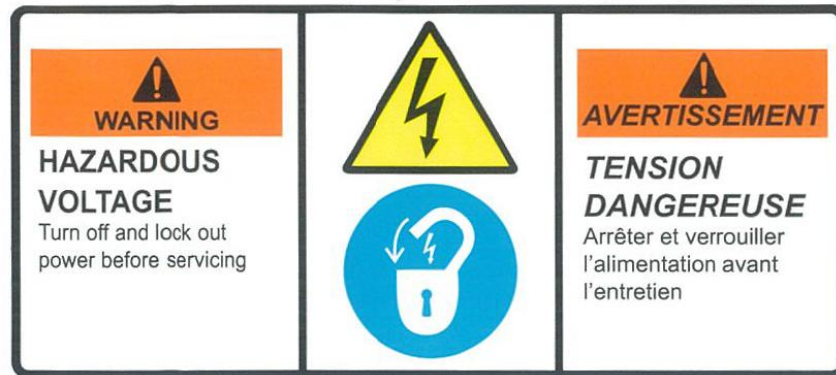


❖ Mechanical Outer Barrier Shutter Plate Movement Signs

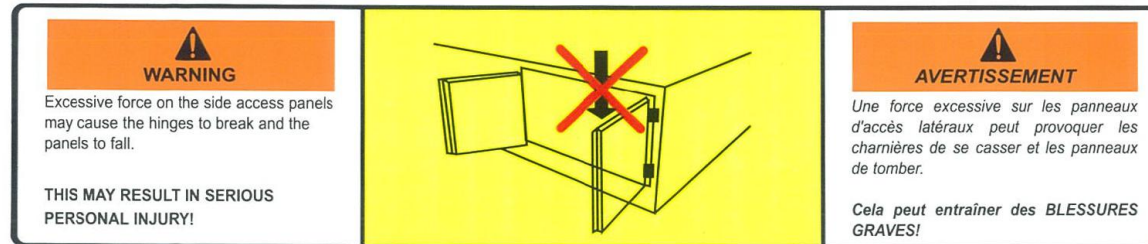
Allocated: Outer Barrier (Left and Right)

ABI System Warning / Caution Labels

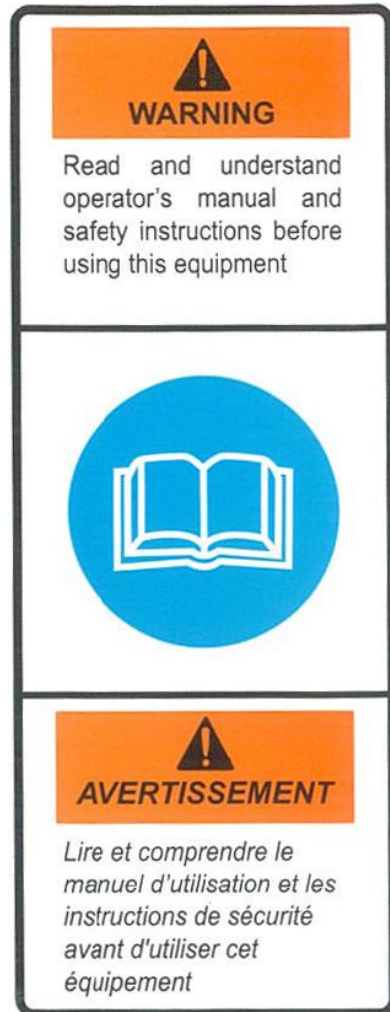
Automated Board Inspection



- ❖ **Electrical Hazard and Lock/Tag Out Signs**
Allocated: PDU



- ❖ **Avoid Excessive Force on Service Door Signs**
Allocated: All services doors



- ❖ **Only Allow Certify User Signs**
Allocated: Front Service Door (Left Side)



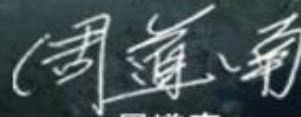
- ❖ **X-ray On/Radiation Signs**
Allocated: Front Service Door



WEE KAH KHIM
Vice President & General Manager



GARY LEONG
Sales Director



周道南
Sales General Manager
(China & Taiwan)



SEOW ZI YANG
Business Development
Manager



FAHMI HELMI
Business Development
Manager